

Entanglement Topology and Noise Determine Quantum Advantage in N -Player Trading Games: Simulation Landscape and Hardware Validation

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Abstract—We extend the two-player quantum Hawk–Dove trading game of Khan et al. to N players under five entanglement topologies (GHZ, W, ring, star, fully connected) using the Eisert–Wilkens–Lewenstein (EWL) protocol, and map how topology, player count, and depolarizing noise jointly determine quantum advantage. Three findings organize the landscape. First, for the GHZ entangler the fixed cooperative profile pays exactly V/N ; other topologies are evaluated cell by cell and may or may not attain that payoff. At GHZ $N=2,3$ the profile passes the restricted $\{D, H, Q_N\}$ deviation check, not an equilibrium test against arbitrary $SU(2)$ gates; ring and fully connected also pass the restricted check at $N=3$ in the fixed-mode artifact. Second, under noise the restricted-menu equilibrium and payoff-gap criteria *disagree*. Matched-count controls reduce gate-budget confounding within the XX family but do not eliminate absolute payoff ordering when zero-noise gaps differ; W is the per-gate-retention outlier. W retains a positive mean circuit-relative payoff gap while hiding position-locked per-player losses. In an exploratory W $N=4$ finite-round pilot, no tested setting produced a beneficial converged fairness repair. Third, on an IBM Heron r2 processor (ibm_fez), a scaling experiment at $N = 3, 4, 5$ on one pinned qubit chain retains 99.4%/98.7%/98.1% of the ideal advantage $3/N$ after readout mitigation and zero-noise extrapolation, and a registered extension reaches $N=7$ at 98.1% of ideal with state fidelity already down to 0.78. A registered 49-pub multi-topology batch puts the topology, equilibrium and fairness claims on device: all twelve runnable topology $\times N$ cells retain 97.9–99.6% of their own ideal advantage; all three unilateral Hawk deviations at GHZ $N=3$ return positive gaps, establishing the restricted-menu equilibrium on hardware; and a wiring permutation that reassigns players to different physical qubits leaves the star $N=4$ hub’s disadvantage intact, so position-locked unfairness is a property of the entanglement topology rather than of qubit allocation. For the registration-source execution, a single depolarizing parameter fitted to the $N=3$ point misses the registered mitigated-fold-1 endpoints at $N=4$ by 0.0043 (2.14σ) and at $N=5$ by 0.0101 (5.17σ); these source-run residuals are retrospective. The same-sign $N=5$ deficit recurs in the prospective repeat, although sign reproduction was not itself a registered test. The CZ-exponential baseline carries the best score of five registered candidates in all three executions at $N \leq 5$, but it misses $N=5$ outside the shared comparison yardstick, and in the registered $N=6,7$ extension both it and the depolarizing model fail their registered tests while the ranking between them inverts at $N=7$; the ranking is not a physical scaling law and does not extrapolate. The mean-payoff observable is first-order insensitive to single bit-flips, so the advantage decays several times slower than state fidelity — a structural robustness with direct consequences for quantum market design.

Index Terms—quantum games, Nash equilibrium, EWL pro-

ocol, entanglement topology, NISQ, error mitigation, quantum finance

I. INTRODUCTION

Financial markets are coordination problems with adversarial payoffs. When traders compete over a scarce asset, each chooses between an aggressive position — capturing the full value if unopposed — and a cooperative one that shares value reliably. The Hawk–Dove game formalizes this tension. Under the live convention used throughout this work, $V=4$ and $C=3$, Hawk strictly dominates Dove in the two-player payoff matrix: it pays more against either opposing action, so mutual Hawk is the unique classical pure Nash equilibrium. Yet mutual Dove pays each player $V/2 = 2$, whereas mutual Hawk pays only $(V - C)/2 = 0.5$. The resulting gap isolates the mechanism-design problem: individually rational aggression destroys value that cooperation would preserve. Classical remedies are external to the game, including trust between counterparties, a trusted intermediary, or regulation that taxes defection.

Quantum game theory offers a different lever. In the Eisert–Wilkens–Lewenstein (EWL) protocol [1], strategies live in $SU(2)$ and players’ qubits are entangled before any move is made; the correlation structure of the shared state then changes what is self-enforcing. The celebrated two-player result is a quantum Nash equilibrium that is simultaneously symmetric and Pareto-optimal: cooperation pays the same as mutual trust but is enforced by interference rather than by an institution. For market mechanisms — carbon-trading schemes, order-book architectures, DeFi protocols — this suggests replacing a regulatory cost with a physical correlation.

Khan et al. [2] demonstrated a quantum Nash-equilibrium advantage for the two-player trading game on ion-trap hardware. Two players, however, admit exactly one entanglement geometry. Real markets are networks: the *topology* of who is correlated with whom is a design variable, and each topology induces a different interference structure, different equilibria, and a different response to decoherence. Varsamis et al. [3] studied N -player entanglement operators and scalability on IBM hardware; our contribution is orthogonal: we fix a financially interpreted game and chart how *topology $\times$ player count $\times$ noise* determine both the magnitude and the equilibrium

status of the quantum advantage, then test the resulting noise model against registered predictions on hardware.

Our contributions:

- 1) **Advantage landscape.** The complete $(N, \text{topology})$ circuit-relative payoff-gap map for $N = 2\text{--}6$ over five topologies. For GHZ, the fixed cooperative profile $(Q_N, \dots, Q_N)$ pays exactly V/N and passes the restricted $\{D, H, Q_N\}$ deviation check at $N = 2, 3$, not against arbitrary $SU(2)$ gates; ring and fully connected also pass that restricted check at $N = 3$. Other topology cells may or may not attain V/N .
- 2) **Noise-robustness dichotomy.** Under depolarizing noise the restricted-menu equilibrium criterion and circuit-relative payoff-gap criterion order topologies differently; noise also breaks player symmetry, producing position-locked per-player disadvantage that follows the entangler's gate ordering (shown by a wiring-permutation experiment). Matched-gate-count and compilation controls reduce gate-budget confounding within the XX-product family (GHZ/ring/star/fully connected), whose normalized decay is similar in the control table, while absolute GHZ-W ordering can remain because their zero-noise payoff gaps differ. W is the normalized-retention outlier at approximately 0.8–1.0 per CX. In an exploratory $N=4$ finite-round pilot, no tested setting produced a beneficial converged fairness repair.
- 3) **Predictive hardware validation.** An $N = 3, 4, 5$ scaling experiment on `ibm_fez` (one pinned chain, one batch job) with tensored readout mitigation and cz -fold zero-noise extrapolation retains 98–99% of the ideal advantage $3/N$; the one-parameter depolarizing fit misses the retrospective source-run mitigated-fold-1 endpoints at $N=4$ by 0.0043 (2.14σ) and $N=5$ by 0.0101 (5.17σ). In the prospective repeat the conditional $N=4$ endpoint passes and the $N=5$ deficit repeats, observationally rather than as a registered sign test. The CZ-exponential baseline scores lowest among the registered candidates but still misses $N=5$ outside the shared ranking yardstick. We identify the mechanism — first-order insensitivity of the mean payoff to single bit-flips — that makes the advantage more noise-robust than the state.
- 4) **Reproducible pipeline.** An open-source harness with circuit-identity gates, a mandatory device-model dress rehearsal before any hardware submission, and crash-safe job recovery.

II. BACKGROUND AND RELATED WORK

A. The Hawk–Dove trading game

Two traders compete over a single asset of value V ; conflict costs C . Hawk is the aggressive position (a short: it captures the full value if unopposed but destroys value in symmetric conflict); Dove is the cooperative position (a long: it shares value reliably). The payoff matrix is

	opp. Dove	opp. Hawk
you Dove	$V/2, V/2$	$0, V$
you Hawk	$V, 0$	$(V-C)/2, (V-C)/2$

We fix $V=4$, $C=3$ throughout (the repo's live convention, `src/config.py`). At these values Hawk strictly dominates Dove: against Dove it pays $V = 4$ rather than $V/2 = 2$, and against Hawk it pays $(V - C)/2 = 0.5$ rather than 0. The unique classical pure Nash equilibrium is therefore mutual Hawk, even though mutual Dove is Pareto-superior. The familiar interior mixed equilibrium with Hawk probability V/C applies only when $V < C$; here $V/C = 4/3$ lies outside the probability simplex. This distinction is why the paper uses the all-Hawk payoff as its classical pure-equilibrium baseline.

B. The EWL protocol and its N -player extension

The EWL quantization [1] embeds strategies in $SU(2)$ and entangles qubits before strategy application:

$$|\psi_{\text{out}}\rangle = J^\dagger \left(\bigotimes_{j=1}^N U_j \right) J |0\rangle^{\otimes N}, \quad J = e^{i\frac{\gamma}{2} X^{\otimes N}}, \quad (1)$$

with maximal entanglement at $\gamma = \pi/2$ [4], [5]. The N -appropriate quantum strategy generalizes to $Q_N = U(0, \pi/N, \pi/N)$; at $\gamma = \pi/2$ the profile $(Q_N, \dots, Q_N)$ maps $|0\rangle^{\otimes N}$ back to $|0\rangle^{\otimes N}$ exactly, so the ideal output is a single computational basis state at every N .

C. Relation to prior N -player hardware work

Varsamis et al. [3] study N -player entanglement operators and their scalability on IBM hardware generically — characterising the operators themselves across devices. We instead fix a single game (the financial Hawk–Dove) and map its *advantage landscape*: the topology-dependent equilibrium/payoff dichotomy, the noise-induced position-locked unfairness, and a registered predictive noise model whose out-of-sample predictions we test on hardware. Chappell, Iqbal, and Abbott [5] compared GHZ and W entanglement in N -player quantum games in an EPR setting, but in simulation and without a financial payoff structure or a hardware validation; our contribution is to carry that GHZ-versus-W question through a concrete trading game to a noise-model-tested result on a real device.

III. MODEL AND METHODS

A. N -player Hawk–Dove payoffs

For N players the payoff generalises as a Benjamin–Hayden tensor [4]: let k of the N players play Hawk. A Hawk player receives V/k for $0 < k < N$ and $(V-C)/N$ when $k = N$ (all Hawk); a Dove player receives V/N when $k = 0$ (all Dove) and 0 otherwise. The best classical pure Nash equilibrium is all-Hawk, paying $1/N$ per player at $V=4$, $C=3$. For the GHZ fixed profile $(Q_N, \dots, Q_N)$, the analytic GHZ cooperative outcome pays exactly V/N ; its ideal analytic-baseline advantage is therefore $(V-1)/N = 3/N$, a constant $3\times$ the classical payoff. This exact identity is scoped to that GHZ outcome; other topologies are evaluated cell by cell. The full

tensor (a per-player payoff for each of the 2^N pure outcomes) feeds the best-response Nash analysis of Sec. IV.

B. Entanglement topologies

We study five entanglement patterns with illustrative, non-one-to-one market analogies: **GHZ** (maximal all-to-all correlation; a fully centralised order book), **W** (single-excitation entanglement robust to one-qubit loss; a partial-information market), **ring** (each player entangled with its two neighbours; a circular trading pit), **star** (a central hub connected to all leaves; a market-maker / dealer model), and **fully-connected** (every pair entangled; a dense interbank network). Each entangler is realised gate-natively in the pinned $\{u, cx\}$ basis: GHZ as a Hadamard-conjugated CNOT-ladder parity rotation ($2(N-1)$ CX per J), W via the exact prep-conjugated construction (a prep cascade + anti-controlled RX; T8 spec), and the pairwise topologies (ring, star, fully-connected) via one $R_{XX}(-\gamma)$ per edge.

C. Payoff-gap metrics and honesty conventions

The simulation artifacts report a *circuit-relative payoff gap*: the mean payoff of the fixed profile $(Q_N, \dots, Q_N)$ minus the highest-mean pure Nash payoff found in the restricted $\{D, H\}^N$ game after those classical labels pass through the same entangler and, where applicable, the same noisy circuit. This comparator is topology- and noise-dependent. The hardware sections instead report *analytic-baseline advantage*: measured mean cooperative-profile payoff minus the noiseless all-Hawk payoff $(V - C)/N = 1/N$ of the original classical game. We keep the two quantities separate and do not compare their zero crossings directly.

Equilibrium status is also restricted: “Nash” below means no profitable unilateral deviation within the finite menu $\{D, H, Q_N\}$, not equilibrium against arbitrary $SU(2)$ gates. Under noise the mean over players can hide per-player losses; we therefore report per-player vectors alongside means. Per-player hardware breakdowns are diagnostic analyses added after registration; they are not part of the registered acceptance test.

D. Noise model and simulation paths

Noise is a depolarizing channel of strength p applied to every gate in the pinned $\{u, cx\}$ basis — p on each single-qubit u and p on each two-qubit cx (two-qubit ratio $r=1$). Noisy expectation values are computed by exact density-matrix simulation on the gate-level circuits (all five topologies use their exact gate-level entanglers, W via the T8 construction); noiseless reference payoffs use a statevector path. The noiseless payoff tensor and Nash analysis run on the statevector path.

IV. SIMULATION RESULTS: THE ADVANTAGE LANDSCAPE

A. Scaling and topology at zero noise

For GHZ, the fixed profile $(Q_N, \dots, Q_N)$ pays exactly $V/N = 4/N$ per player; its circuit-evaluated restricted $\{D, H\}$ comparator is $1/N$, giving a circuit-relative payoff

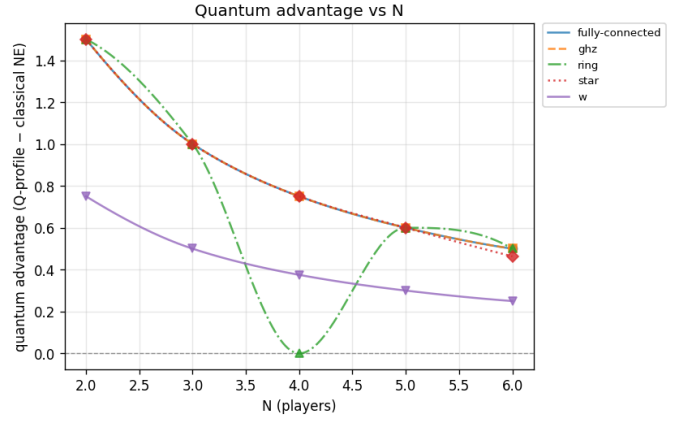


Fig. 1. Zero-noise circuit-relative payoff gap of the fixed profile $(Q_N, \dots, Q_N)$, defined as its mean payoff minus the highest-mean circuit-evaluated restricted $\{D, H\}$ equilibrium payoff, at $\gamma = \pi/2$ and $V=4, C=3$. The $3/N$ gap (star markers) is attained by GHZ and fully connected at every recorded N ; the ring has zero gap at $N=4$; the star shows a weaker $N=6$ value (0.463 vs. 0.5); and W tracks below throughout. Filled markers indicate that the profile passes the restricted $\{D, H, Q_N\}$ deviation check, which depends on topology.

gap of $3/N$ (Fig. 1). The other topologies are evaluated cell by cell. Fully connected attains V/N for all recorded $N = 2-6$ cells; ring does so except at $N=4$; W attains V/N but has a larger circuit comparator and therefore a smaller gap; and star attains V/N through $N=5$ but deviates at $N=6$, with asymmetric player payoffs from $N=4$ onward. The restricted-menu equilibrium status is likewise cell-specific: GHZ passes the $\{D, H, Q_N\}$ deviation check at $N=2, 3$, and ring and fully connected also pass at $N=3$ in the fixed-mode artifact.

For GHZ, a separate $N = 2-8$ computation recorded in docs/VERIFIED-FACTS.md finds that a unilateral Hawk deviation pays $2 + 2 \cos(2\pi/N)$ and that the restricted deviation result agrees with $2 + 2 \cos(2\pi/N) \leq V/N$ in every computed cell. Thus at $N=4$ the deviation pays 2.0 against the cooperative 1.0, and at $N=5$ it pays $\varphi^2 \approx 2.618$ against 0.8. For $N \geq 4$ we therefore report the GHZ profile as a *fixed cooperative protocol*, not as an equilibrium prediction. The landscape also contains an $N=4$ ring interference zero and a star $N=6$ deviation from V/N ; for star the circuit-relative payoff gap is asymmetric across players — at $N=4$ the three leaves gain 1.0 each while the hub gains nothing.

Sweeping γ separates a cooperative payoff gap from an entanglement-enabled restricted-menu equilibrium. At $\gamma = 0$, $J = I$ and the phase-only Q_N profile is measurement-equivalent to all-Dove: its positive payoff gap over all-Hawk is classical in outcome probabilities and the profile does not pass the restricted deviation check. At nonzero γ , the same sweep records when the fixed profile begins to pass that finite-menu check.

At $N=2$ every topology’s circuit-relative payoff gap is positive from $\gamma=0$, and (Q, Q) begins to pass the restricted deviation check from $\gamma = 0.3\pi$ (W excepted, which never passes in range). At $N=3$ GHZ, fully connected, and ring hold

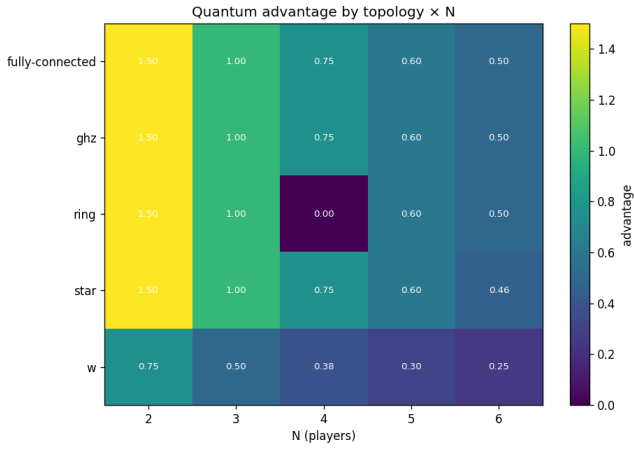


Fig. 2. Circuit-relative payoff-gap heatmap across the $(N, \text{topology})$ grid at $V=4, C=3$ (mean per-player payoff of $(Q_N, \dots, Q_N)$ minus the highest-mean circuit-evaluated restricted $\{D, H\}$ equilibrium payoff). GHZ and fully connected hold the $3/N$ gap everywhere; the single dark cell is the ring at $N=4$ (gap exactly 0).

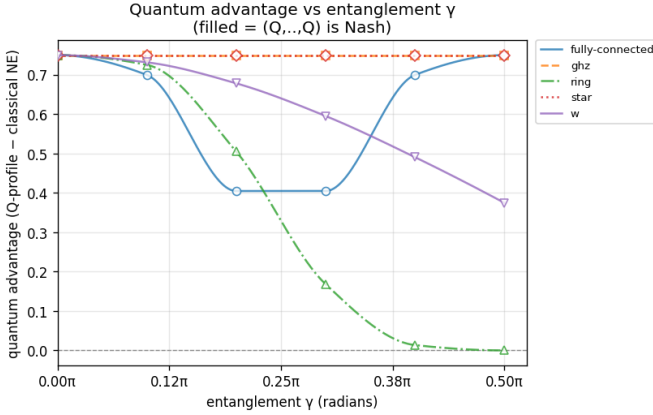


Fig. 3. Circuit-relative payoff gap versus entanglement angle γ at $N=4$ ($V=4, C=3$, noiseless). GHZ and star hold 0.75 flat; fully connected dips and recovers; W decays monotonically; and ring falls to zero at $\gamma = \pi/2$. Data: results/gamma-sweep/N4.

a gap of 1.0 across the sweep and begin to pass at $\gamma = 0.3\pi - 0.4\pi$; star and W never pass. At $N=4$ (Fig. 3) GHZ and star hold 0.75 at every γ , fully connected dips to 0.40 mid-sweep and recovers, W decays monotonically to 0.375, and ring reaches exactly zero at $\gamma = \pi/2$, the same $N=4$ interference zero. No series has a negative circuit-relative payoff gap; only that ring endpoint reaches zero.

The star is the only topology that breaks player symmetry at zero noise. For $N \geq 4$ its per-player circuit-relative payoff gaps are uneven (Fig. 4): at $N=4$ the three leaves have gap 1.0 each while the hub has 0.0; by $N=6$ the hub has 0.952 against the leaves’ 0.366. The mean circuit-relative payoff gap stays close to $3/N$ until $N=6$, where the hub-leaf split pulls it to 0.463.

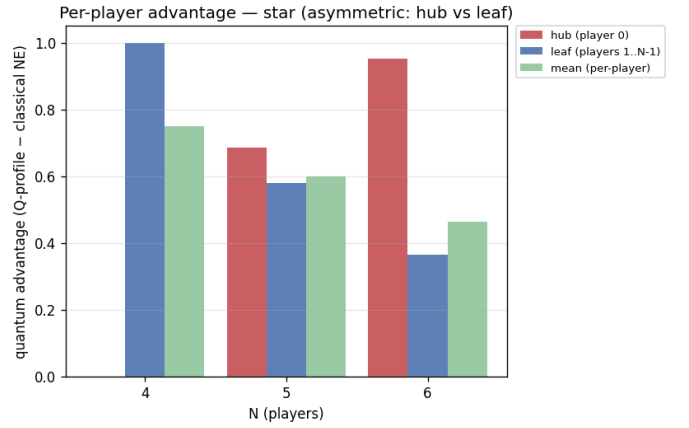


Fig. 4. Star-topology per-player circuit-relative payoff gap (hub player 0 vs. each leaf) at $\gamma = \pi/2$, $V=4, C=3$, noiseless, for $N = 2 \dots 6$. The leaves capture the gap at small N ; the hub captures it at large N ; the mean (dashed) tracks $3/N$ until $N=6$. Data: results/n-scaling-advantage/2026-07-19T0901Z.

B. Noise robustness: two criteria disagree

Sweeping the depolarizing strength p from 0 to 0.05 on the gate-level noisy path, two natural “robustness” criteria give different orderings, so we state which we mean. Under the *restricted-menu equilibrium* criterion, the GHZ cooperative profile passes at $p = 0$ only for $N=2, 3$, and at $N=3$ it loses that status between $p = 0.04$ and 0.045 . For $N=4, 5$ the GHZ thresholds $p^* \approx 0.0197$ and 0.0098 are *circuit-relative payoff-gap zero crossings*, not equilibrium losses: the profile does not pass the restricted deviation check there even at $p = 0$. The fixed-mode zero-noise artifact records ring as passing at $N=2, 3$, fully connected at $N=2, 3$, star at $N=2$, and W at neither $N=2$ nor $N=3$.

Under the *circuit-relative payoff-gap magnitude* criterion the picture flips: at $N=3$ GHZ retains the largest mean gap at $p = 0.05$ (0.333, 33% of its noiseless value) versus ring 0.200 and W 0.047. These are not crossings relative to the analytic $1/N$ baseline: noise can change both the fixed-profile payoff and the highest-mean restricted $\{D, H\}$ circuit equilibrium, and comparator drift can drive a zero crossing. The natural-budget ordering therefore characterizes the production circuits. Matched-count controls reduce gate-budget confounding within the XX family, whose normalized per- cx decay rates are similar in the control table at approximately 1.8–2.7; no statistical equivalence test was recorded. Absolute GHZ–W payoff-gap ordering can remain after those controls because their zero-noise gaps differ. W is the normalized-retention outlier at approximately 0.8–1.0 per CX. It keeps a positive but small mean circuit-relative gap across the whole grid, while the mean hides per-player losses: at $N=5$ two of the five players have negative per-player gaps from $p = 0.01$ (worst -0.077 at $p = 0.015$). All entanglers here are exact gate-level circuits; W uses the T8 construction (44 and 68 cx per J at $N=4, 5$, versus 100/444 for the retired transpiled-unitary fallback).

C. Position-locked unfairness and the adaptation pilot

The noise-induced per-player asymmetry is not a property of the player labels or the ideal state. A wiring-permutation test on the ring at $N=5$ (`scripts/asymmetry_diagnostics.py`) cyclically shifts the qubit-to-player wiring and the whole per-player payoff vector permutes rigidly with it (deviation $\sim 10^{-16}$); and applying the same depolarizing weight to the *ideal final state* instead of per-gate collapses the per-player spread to ~ 0 . The disadvantage therefore follows the entangler’s *circuit position* (gate ordering), not the ideal state or the player identity.

This is now confirmed on hardware. A registered wiring-permutation control on the star at $N=4$ (Sec. V-E, Fig. 7C) reassigns players to different physical qubits and moves the measured per-player payoff vector by at most 0.035, inside the registered threshold, while the hub–leaf split itself (≈ 0.34 against ≈ 1.21) is unchanged. The simulation result above says the disadvantage is not carried by the player label; the hardware result adds that it is not carried by the *qubit* either. What remains is the graph position.

An exploratory W-state pilot at $N=4$ tested one provisional independent round-robin best-response rule. No tested setting produced a beneficial converged fairness repair: at $p = 0$ the dynamics reached the 50-round cap near mean 0.51 without converging; at $p = 0.02$ they entered a payoff limit cycle with lower mean and larger spread; and at $p = 0.05$ they converged with little change. Two robustness checks were then run on this pilot, both simulation-only. Replacing round-robin with *simultaneous* best response, which removes move order entirely, reproduces all three verdicts, so the $p = 0.02$ non-convergence is a property of the noisy best-response map rather than of the update order. Extending to $N=5$, W $N=5$ reproduces the $N=4$ verdicts, while ring $N=5$ fails to converge at every noise level tested and does so in the opposite direction: it drives a perfectly fair $p=0$ baseline apart while leaving mean welfare unchanged, i.e. it redistributes rather than shrinks. Because those ring endpoints are snapshots of non-convergent trajectories we report the direction only and not the magnitudes. Belief-based rules such as fictitious play remain untested, and the pilot remains exploratory pending game-theory sign-off.

V. HARDWARE VALIDATION ON IBM HERON R2

A. Pipeline and safeguards

Every submission passes three gates: (i) an `Operator.equiv` circuit-identity assertion against the dense $J^\dagger(U^{\otimes N})J$ reference, (ii) a noiseless simulator dry-run that must reproduce the exact analytic advantage, and (iii) a full dress rehearsal of the batch — including mitigation analysis — on the device noise model reduced to the executed qubits. Job identifiers are persisted before result polling, making runs crash-safe and recoverable.

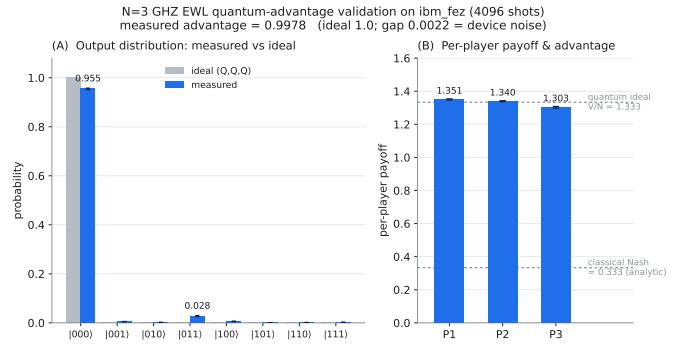


Fig. 5. Hardware validation of the $N=3$ GHZ EWL quantum advantage on `ibm_fez` (IBM Heron r2). The validated (Q, Q, Q) circuit — entangler J decomposed to 6 native cz gates — was executed with 4096 shots. (A) Measured output distribution over the eight computational-basis states (blue) versus the noiseless ideal (grey); error bars are binomial shot noise. 3911/4096 shots (95.5%) fall in $|0\dots 0\rangle$, the ideal (Q, Q, Q) output. (B) Measured per-player payoff (blue) against the analytic classical-Nash baseline (0.333, noiseless) and the quantum ideal $V/N = 1.333$; error bars propagate shot noise. The measured mean payoff 1.331 gives advantage 0.9978 vs the ideal 1.0. Physical qubits $\{136, 143, 142\}$ (calibration 2026-07-15): readout error 0.38–0.40%, cz error 136–143: 0.17%, 143–142: 0.23%.

B. Single-point validation at $N = 3$

We ran the validated Q_3 circuit (entangler decomposed to 6 native cz gates) with 4096 shots on `ibm_fez`; 3911/4096 shots landed in $|000\rangle$ (Fig. 5). The measured mean payoff is 1.3311, or 99.835% of the ideal payoff $4/3$; subtracting the analytic all-Hawk baseline $1/3$ gives measured advantage 0.9978, or 99.780% of the ideal advantage 1.0.

A product of the recorded readout fidelities (0.9960, 0.9961, 0.9962) and two-qubit fidelities (edge errors 0.17% and 0.23%, applied in the recorded four-plus-two CZ split) gives a heuristic ground-state probability of about 0.977, whereas the measured value is $3911/4096 = 0.955$. Under a fixed binomial interpretation the difference is about 9.5 standard deviations, so the product has the right order of magnitude but substantially underpredicts non-ground weight; it is not a calibrated statistical model of this execution.

For every outcome with $0 < k < N$ Hawks, total payoff remains V , so the mean remains V/N . Only the all-Hawk outcome lowers total payoff to $V - C$:

$$\bar{\pi} = \frac{V}{N} - \frac{C}{N}P(1\dots 1). \quad (2)$$

In the measured counts, 176 of the 185 shots outside $|000\rangle$ land on one- and two-Hawk outcomes, while only the 9 all-Hawk shots (0.22%) lower the mean. The identity reconstructs the measured payoff 1.3311 and advantage 0.9978. The mean-payoff observable is therefore first-order insensitive to single bit-flips from the ideal outcome: errors that flip one qubit redistribute payoff between players without lowering its mean. This is the structural mechanism behind the scaling section’s central observation — the advantage decays several times slower than the state fidelity — and it is why the $N=3$ advantage survives at 99.780% of its ideal while $P(|000\rangle)$ has already lost 4.5%.

TABLE I

MEASURED ADVANTAGE ON IBM_FEZ FOR THE REGISTRATION-SOURCE EXECUTION (4096 SHOTS). REPEAT EXECUTIONS ARE DISCUSSED SEPARATELY; THE CROSS-CALIBRATION AGGREGATE OVER BOTH EPOCHS IS FIG. 6. IDEAL ADVANTAGE IS $3/N$; RETENTION IS ZNE/IDEAL.

N	ideal	raw	mitig.+ZNE	retention	$P(0\dots 0\rangle)$
3	1.00	0.9919	0.9938	99.4%	0.938
4	0.75	0.7386	0.7403	98.7%	0.922
5	0.60	0.5837	0.5887	98.1%	0.868

C. Scaling at $N = 3, 4, 5$ with error mitigation

All three N ran on prefixes of one pinned 5-qubit chain [59, 75, 74, 73, 79] (chosen by calibration error) in a single 11-pub batch: two readout-calibration circuits plus $N \in \{3, 4, 5\} \times \text{cz-fold } \lambda \in \{1, 3, 5\}$. Readout errors are corrected by tensored confusion-matrix inversion (constrained least squares) [6]; zero-noise extrapolation folds each cz locally ($\text{cz} \rightarrow \text{cz}^\lambda$, unitary-preserving since cz is self-inverse) and extrapolates the readout-mitigated payoff linearly to $\lambda = 0$ [7], [8].

Two model predictions frame Table I: the reduced device noise model (`NoiseModel.from_backend`) and a *fit-one-predict-two* test in the paper’s own depolarizing model, with a single $p_{\text{eff}} = 0.0018$ fitted to the $N=3$ mitigated-fold-1 point. For the registration-source execution, the registered primary mitigated-fold-1 endpoint misses at $N = 4$ by 0.0043 (2.14σ) and at $N = 5$ by 0.0101 (5.17σ). These source-run residuals are retrospective because the registration was written after that execution. For the displayed ZNE endpoint, the corresponding misses are 0.0041 (2.07σ) and 0.0060 (3.05σ). The second execution is the prospective registered repeat: its conditional $N = 4$ endpoint passes and its $N = 5$ endpoint misses by 0.0092 (4.74σ). The repeated deficit sign is observational, not a registered sign test.

The CZ-exponential baseline carries the smallest mean $|z|$ — the best score — of the five registered candidates in all four executions, including both cross-calibration repeats. It nevertheless misses $N = 5$ outside the shared 95% comparison yardstick in every run; that yardstick is for ranking, not a model-specific calibrated interval. **That ranking is specific to $N \leq 5$:** in the registered $N = 6, 7$ extension (Sec. V-D) the CZ-exponential law loses to the depolarizing model at $N=7$ and both fail their registered tests, so the ranking is not a physical scaling law and does not extrapolate. The registered target of three to five distinct calibrations is now met at three, and the $N = 5$ deficit reproduces with the same sign in all three repeats. The deficit magnitude is not stable across epochs, however: it ranges from 0.0092 to 0.0186, so the epoch-to-epoch variation is itself of order the effect. Model-specific predictive uncertainties remain needed, and the cross-calibration sample that is chain-matched is two epochs rather than three.

D. Extension to $N = 6, 7$

A later execution extends the curve to $N = 3, \dots, 7$ on one pinned seven-qubit chain [97, 107, 108, 109, 110, 111, 98] (17 pubs, 20 QPU-seconds). The chain was written into the registration and held at submission, so the executed qubits provably match the ones the predictions describe. Measured mitigated fold-1 advantage retains 99.2%, 97.5%, 97.3%, 96.3% and 98.1% of the ideal $3/N$ at $N = 3, \dots, 7$ while $P(|0\dots 0\rangle)$ falls from 0.943 to 0.784: **the advantage survives to seven players at 98% of ideal with the state fidelity already down to 0.78**, which is the first-order-insensitivity mechanism operating across the full curve.

Two competing one-parameter models were registered before submission, both fitted on the same $N=3$ point and extrapolated with no $N > 3$ information: the depolarizing p_{eff} model and the CZ-exponential retention law $A_N = A_N^{\text{ideal}} r^{\text{cz}N}$. *Both failed the registered 1.96σ test at $N = 6, 7$, and they failed differently:* the depolarizing model overpredicts at every N , while the CZ-exponential model overpredicts at $N = 4, 5, 6$ and then under-predicts at $N = 7$. A one-parameter retention law fitted at $N=3$ cannot reproduce that curvature.

The registered model race also split: the CZ-exponential law is closer at $N = 4, 5, 6$ but the depolarizing model is closer at $N = 7$. **The CZ-exponential ranking reported below therefore does not extrapolate past $N=5$,** and we scope it accordingly. Finally, measured advantage is strictly decreasing in N (the registered monotonicity check), but *retention* is not: $N=7$ keeps more of its ideal advantage than $N = 4, 5, 6$ despite carrying the most gates and the lowest fidelity, and its ZNE estimate overshoots the noiseless ideal (0.4437 against 0.4286), which no physical noise model can produce. We report this as an unexplained observation rather than a result. Distinguishing payoff geometry at odd N from an end-of-chain qubit effect or an extrapolation artefact requires a repeat of *this $N = 3 \dots 7$ batch specifically*: the cross-calibration repeats aggregated in Fig. 6 re-execute the registered $N = 3, 4, 5$ batch and so cannot address it, and no such $N = 6, 7$ repeat has been run.

E. Topology, equilibrium and fairness on device

The experiments above are all GHZ. To put the paper’s topology, equilibrium and fairness claims on device data we ran a single 49-pub batch (4096 shots, 55 QPU-seconds) spanning five entangler topologies, unilateral strategy deviations, wiring permutations and an entanglement-angle sweep, with all predictions registered before submission. Cells were selected from *measured* routed CZ counts on the real coupling map rather than from pre-routing gate counts: on heavy-hex hardware the two differ sharply, and fully-connected $N=5$ (67 CZ) and W $N=4, 5$ (107, 219) were excluded on that basis.

Every cell retains 97.9–99.6% of its own ideal advantage (Table II, Fig. 7A), including W $N=3$ at 42 CZ — seven times the gate count of GHZ $N=3$. The mechanism therefore is not a GHZ artefact: ring $N=5$ and star $N=5$ sit at $P(|0\dots 0\rangle) \approx 0.10\text{--}0.11$, states that are wrecked by any fidelity measure,

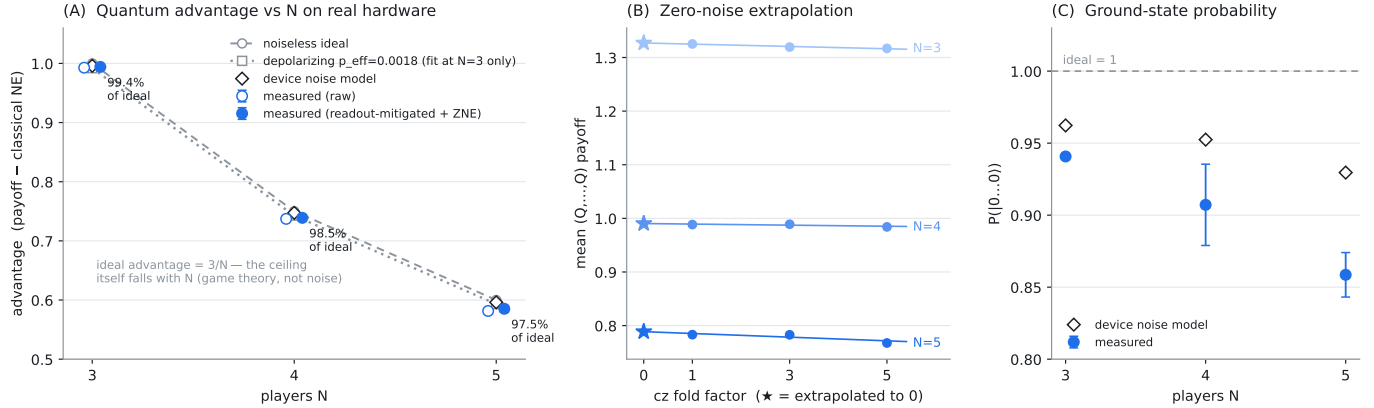


Fig. 6. Cross-calibration aggregate of the N -player GHZ EWL quantum advantage on `ibm_fez` (IBM Heron r2): three executions spanning *two distinct provider calibrations*, all on the same physical chain [59, 75, 74, 73, 79], 4096 shots each. Executions sharing a calibration stamp are averaged into one point before the spread is taken, because two runs under one calibration are one sample rather than two; the two epochs are 2026-07-16 08:30:13 (two executions) and 2026-07-26 13:29:01 (one). A fourth execution exists on chain [20, 21, 22, 23, 24] and is deliberately *excluded* here: pooling it would confound calibration drift with a change of physical qubits. Table I reports the registration-source execution alone. (A) Measured cooperative-profile advantage (mean $(Q, \dots, Q)$ payoff minus the analytic noiseless classical-Nash payoff) at $N = 3, 4, 5$: raw (open) and readout-mitigated + ZNE (filled), against the noiseless ideal (1.0, 0.75, 0.6), the device noise-model prediction (diamonds), and a single-parameter depolarizing model with $p_{\text{eff}} = 0.0018$ fitted to the $N=3$ point alone — its $N=4, 5$ values are predictions, not fits. The ideal, device-model and depolarizing curves are the registration anchor’s rather than run averages, so the registered p_{eff} is quoted unchanged. Error bars are the sample standard deviation across the two calibration epochs — a genuine cross-calibration estimate, unlike the within-epoch spread reported in earlier drafts — and still exclude the within-run multinomial and weighted-fit standard errors, so they are a lower bound on total uncertainty. (B) ZNE mechanics: readout-mitigated payoff vs cz fold factor ($\lambda = 1, 3, 5$) with the weighted linear fit extrapolated to $\lambda = 0$ (stars). (C) Ground-state probability $P(|0 \dots 0\rangle)$: the state fidelity decays several times faster than the advantage because the mean-payoff observable is first-order insensitive to single bit-flips from $|0 \dots 0\rangle$. $(Q, \dots, Q)$ passes the restricted-menu pure-Nash check only at $N=3$ among the plotted genuinely multiplayer cases; at $N=4, 5$ the profile is the cooperative quantum protocol, not an equilibrium.

TABLE II

MULTI-TOPOLOGY BATCH ON `IBM_FEZ` (JOB `D9IA1PD0K0JC738JAQGG`, 4096 SHOTS, ONE PINNED QUBIT SET). RETENTION IS MEASURED MITIGATED FOLD-1 ADVANTAGE AS A FRACTION OF *that cell’s own* NOISELESS ADVANTAGE, WHICH IS NOT $3/N$ FOR EVERY TOPOLOGY.

topology	N	cz	ideal adv	measured	retention
GHZ	3	6	1.0000	0.9886	98.9%
GHZ	4	9	0.7500	0.7425	99.0%
GHZ	5	14	0.6000	0.5921	98.7%
ring	3	10	1.0000	0.9899	99.0%
ring	4	20	0.0000	0.0894	—
ring	5	28	0.6000	0.5960	99.3%
star	3	5	1.0000	0.9934	99.3%
star	4	9	0.7500	0.7423	99.0%
star	5	17	0.6000	0.5977	99.6%
fully-conn.	3	10	1.0000	0.9887	98.9%
fully-conn.	4	31	0.7500	0.7389	98.5%
W	3	42	1.0000	0.9792	97.9%

and still retain over 99% of their advantage. The topology axis does not separate these systems by how much advantage they retain; it separates them by *structure*, as the next two results show.

a) The restricted-menu equilibrium holds on hardware:

At GHZ $N=3$ the three unilateral Hawk deviation gaps measure $+0.2895$, $+0.2726$ and $+0.3512$ (Fig. 7B), all positive and many standard errors clear of zero. This was a registered test and it passed at every deviating position: the (Q, Q, Q) profile is a pure Nash equilibrium in the restricted menu

$\{D, H, Q_N\}$ on the device, not only in simulation. It remains *not* a full-SU(2) equilibrium claim.

A registered sweep of the same gap in the entanglement angle changes sign as predicted: -0.6877 at $\gamma = 0.30\pi$ and $+0.2467$ at $\gamma = 0.45\pi$. The intermediate point $\gamma = 0.40\pi$ was registered in advance as marginal and excluded from the pass criterion; it returned $+0.0067$, statistically indistinguishable from zero. Entanglement strength is therefore what converts the cooperative profile into an equilibrium, measured on hardware, with the transition bracketed near $\gamma = 0.4\pi$ at $N=3$.

b) Position-locked unfairness is a property of the topology: star $N=4$ is maximally unequal even noiselessly: the analytic per-player vector is $(0.25, 1.25, 1.25, 1.25)$, the hub being the loser. On device the measured vector is $(0.353, 1.202, 1.212, 1.202)$, and under two wiring permutations that reassign players to different physical qubits it moves by at most 0.035 (Fig. 7C) — inside the registered threshold. The registered null therefore holds: **the disadvantaged position is determined by the entanglement topology, not by which physical qubit a player was allocated**. This is the hardware control that the position-locked unfairness claim of Sec. IV-B previously lacked.

c) A cell whose advantage is made of noise: ring $N=4$ is the one cell whose noiseless circuit is deterministic on $|1 \dots 1\rangle$ — the all-Hawk outcome — so its ideal advantage is exactly zero. It measured $+0.089$, the smallest of the twelve cells but clearly positive, and zero-noise extrapolation pulled it back to

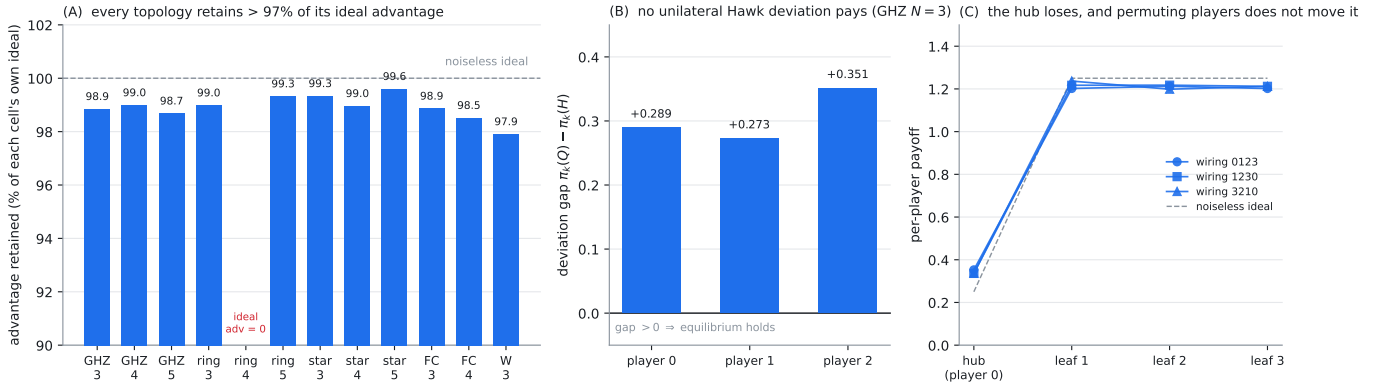


Fig. 7. Multi-topology EWL batch on `ibm_fez` (job `d9ia1pd0k0jc738jaqgg`), 49 pubs $\times$ 4096 shots on one pinned qubit set, all predictions registered before submission. (A) Measured mitigated fold-1 advantage as a percentage of each cell’s *own* noiseless advantage; every cell retains 97.9–99.6%. ring $N=4$ carries no percentage because its noiseless advantage is exactly zero — its ideal circuit is deterministic on $|1\dots 1\rangle$, so the +0.089 it measures is created by decoherence, and zero-noise extrapolation correctly pulls it back toward zero (+0.037). (B) Unilateral Hawk deviation gaps $\pi_k(Q, Q, Q) - \pi_k(H \text{ at } k)$ at GHZ $N=3$: all three are positive, so no deviation pays and the restricted-menu equilibrium holds on hardware. (C) star $N=4$ per-player payoffs under three wirings. The hub earns ≈ 0.34 while each leaf earns ≈ 1.21 , and permuting players across physical qubits does not move the pattern: the disadvantage is locked to the graph position, not to the qubit.

+0.037. All three parts of that were registered in advance. A pipeline that merely inflated whatever it touched could not produce a cell where decoherence *creates* apparent advantage and the extrapolation correctly removes it.

d) *What did not work*: The registered per-cell predictions of *absolute* advantage failed, in both the raw device-noise-model form and a CZ-linear bias-corrected form, and that comparison is additionally confounded: the provider recalibrated between registration and submission and the qubit selector moved the pinned set, so those z -scores mix model error with a change of qubits. The five results above are unaffected because each is a within-job differential — a payoff difference between two profiles measured on the same qubits in the same job, a permutation against its own twin, or one cell ranked against the other eleven. Absolute-magnitude prediction of NISQ payoffs from a device noise model remains out of reach here; the structural claims do not depend on it.

F. Why the advantage outlives the state

Ground-state probability decays several times faster than the advantage (Table I): a single bit-flip from $|0\dots 0\rangle$ yields a one-Hawk outcome whose *mean* payoff is still exactly V/N , so the mean-payoff observable is first-order insensitive to uncorrelated single flips. Readout mitigation consequently moves the mean very little; its per-player effects are mixed and remain diagnostic analyses rather than preregistered acceptance tests. The displayed ZNE estimates are higher for every series, but no recorded population analysis attributes that shift specifically to suppression of CZ-created all-Hawk weight. The structural robustness is a property of the game’s payoff geometry, not of the device.

The same single-flip terms that leave the mean invariant are what create winners and losers. Under the all-Dove branch each one-Hawk outcome transfers V to a single player and zero to the rest: the mean is protected, the individual player is

not, and under gate-level noise the disadvantaged position is locked to the entangler’s circuit layout rather than to any coordination failure (Sec. IV-B). In the single exploratory W-state $N=4$ finite-round pilot, the provisional independent round-robin best-response rule produced no beneficial converged fairness repair in any tested setting. At $p=0$ the dynamics reached the 50-round cap near mean 0.51 without converging; at $p=0.02$ they entered a payoff limit cycle with lower mean and larger spread; and at $p=0.05$ they converged with little change. The verdict survives both checks we applied to it: it is unchanged under simultaneous rather than round-robin updates, and W $N=5$ reproduces it. Ring $N=5$ does not converge at any noise level tested and fails in the opposite direction, spreading a perfectly fair baseline apart at unchanged mean welfare — reported as a direction, not a magnitude, since those endpoints are non-convergent snapshots. No belief-based rule has been tested. The advantage we report can therefore coexist with unequal per-player outcomes; the per-player floor is a relevant diagnostic for mechanism design.

VI. DISCUSSION

The topology sweep is a market-design map: each entanglement pattern suggests a connectivity architecture, although these analogies are illustrative rather than one-to-one. GHZ represents a centralised order book, star a hub-and-spoke market-maker or dealer model, ring a circular trading pit or interdealer network, and fully connected a dense interbank market. The two-criteria dichotomy of Sec. IV-B yields a concrete design rule: choose the topology by whether the mechanism needs the *equilibrium* or the *payoff*. Restricted-menu equilibrium results are cell-specific: at $N=3$, GHZ, ring, and fully connected all pass the $\{D, H, Q_N\}$ deviation check in the fixed-mode artifact. The noise sweep shows that GHZ’s status is fragile and is lost at finite p ; it does not establish GHZ as uniquely self-enforcing across topologies.

If the goal is a large, durable cooperative payoff (accepting a fixed protocol rather than an equilibrium), GHZ leads at small N on advantage magnitude, while W’s superior per-gate robustness makes it the choice when two-qubit gates — not entanglement structure — are the binding constraint. Two applications illustrate the analogy. In carbon markets (an N -player Hawk–Dove between emission reduction and allowance purchase), the topology that maximises cooperative payoff at a given noise budget is the optimal architecture for a quantum-enhanced trading mechanism. In DeFi, where front-running (MEV) is an N -player Hawk–Dove that destroys value, entangling wallet interactions via a smart-contract-controlled quantum channel could enforce cooperative outcomes without a trusted intermediary — the entanglement graph suggests a possible contract-interaction architecture.

Limitations. Discrete strategy set $\{D, H, Q_N\}$ (mixed/continuous deviations are future work); the fixed GHZ-derived Q_N is not topology-optimized under noise. The chain-matched scaling series uses [59, 75, 74, 73, 79]; the separate $N=3$ validation used [136, 143, 142] and the preceding recorded calibration date. Cross-calibration uncertainty now rests on two chain-matched epochs (three distinct calibrations exist, but one ran on a different physical chain and is excluded from the aggregate), and the epoch-to-epoch variation in the $N=5$ deficit is of order the deficit itself. The T9 adaptation evidence uses one provisional finite-round rule; it has been checked against a simultaneous-update rule and extended to W $N=5$ and ring $N=5$, but belief-based rules such as fictitious play remain untested.

VII. CONCLUSION

We asked whether the two-player quantum Hawk–Dove advantage survives the transition to real N -player networks — topologies, noise, and hardware. The landscape is now charted, and the answer is yes, with sharper edges than the two-player picture suggests. At $N=3$ the GHZ cooperative profile is a genuine symmetric restricted-menu pure Nash equilibrium, and it reproduces on superconducting hardware with measured advantage 0.9978, or 99.780% of the ideal advantage 1.0 (job d9br6166hjag73fg3g7g). At $N=4, 5$ the same profile ceases to be an equilibrium — a unilateral Hawk deviation profits — yet its cooperative payoff retains a large, measurable advantage over the analytic classical baseline on the same device (0.739/0.584 of the ideal 0.75/0.60). The dichotomy — equilibrium at $N=3$, cooperative profile beyond — is real, and it is a feature of the game, not a defect of the experiment. The $N=3$ equilibrium is now established on device rather than inferred from it: a registered test of all three unilateral Hawk deviations returned positive gaps (+0.2895, +0.2726, +0.3512), so no deviation pays on hardware. The curve also extends to $N=7$, where the cooperative advantage still retains 98.1% of its ideal while state fidelity has fallen to 0.78.

Three structural findings carry the design weight. First, the mean-payoff observable is first-order insensitive to single bit-flips, so the advantage decays several times slower than

the state fidelity: quantum market mechanisms degrade gracefully on NISQ devices, and this holds across five entangler topologies, not only GHZ — ring and star at $N=5$ sit at $P(|0\dots 0\rangle) \approx 0.10$ and still retain over 99% of their advantage. Second, the same flip terms that spare the mean concentrate loss on individual players, and that loss is locked to the *graph position*: a registered wiring permutation on star $N=4$ reassigns players to different physical qubits and leaves the hub’s disadvantage intact. In an exploratory finite-round adaptation pilot, the provisional rule reached the 50-round cap without convergence at $p=0$, and no tested setting produced a beneficial converged fairness repair — a verdict unchanged under simultaneous updates and reproduced at W $N=5$, while ring $N=5$ fails to converge at any tested noise level and spreads a fair baseline apart instead of repairing it. Third, the CZ-exponential baseline carries the best registered score in all three scaling executions at $N \leq 5$, but its $N=5$ misses and the shared ranking yardstick do not establish a physical scaling law — and in the registered $N=6, 7$ extension it loses to the depolarizing model at $N=7$ while both models fail their registered tests. The ranking does not extrapolate.

The limitations are the same honesty notes threaded through the text: the $N=4, 5$ curve measures a fixed cooperative protocol, not an equilibrium; the registered target of three to five distinct calibrations is met at three, but only two of those are chain-matched, so the plotted cross-calibration uncertainty rests on two epochs and the $N=5$ deficit magnitude varies by a factor of two across them; the adaptation evidence remains exploratory and finite-round, though its two self-raised caveats — move order, and coverage beyond W $N=4$ — have both been tested without changing the verdict; and *no* registered model predicted absolute advantage magnitudes successfully at any N , so the quantitative noise models here are descriptive rather than predictive. The structural claims above are differential measurements and do not depend on them. Within those bounds the design implication is concrete — entanglement topology is a market-design variable whose noise cost is budgeted per two-qubit gate, and its benefit should be read at the per-player floor.

REPRODUCIBILITY

All circuits, harnesses, per-run calibration snapshots, and analysis code are open source at <https://github.com/Prithvi8706/Multiplayer-QHD-Game> (pinned qiskit 1.3.2 / qiskit-aer 0.14.2 / qiskit-ibm-runtime 0.36.1); every hardware artifact records its job id, chain, and calibration.

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